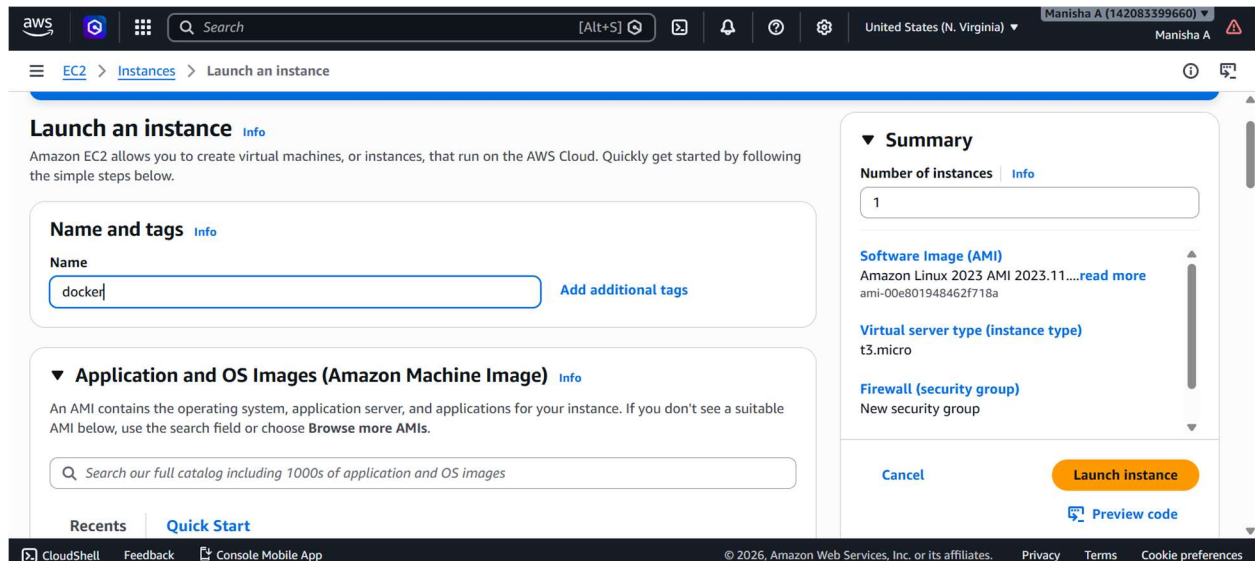


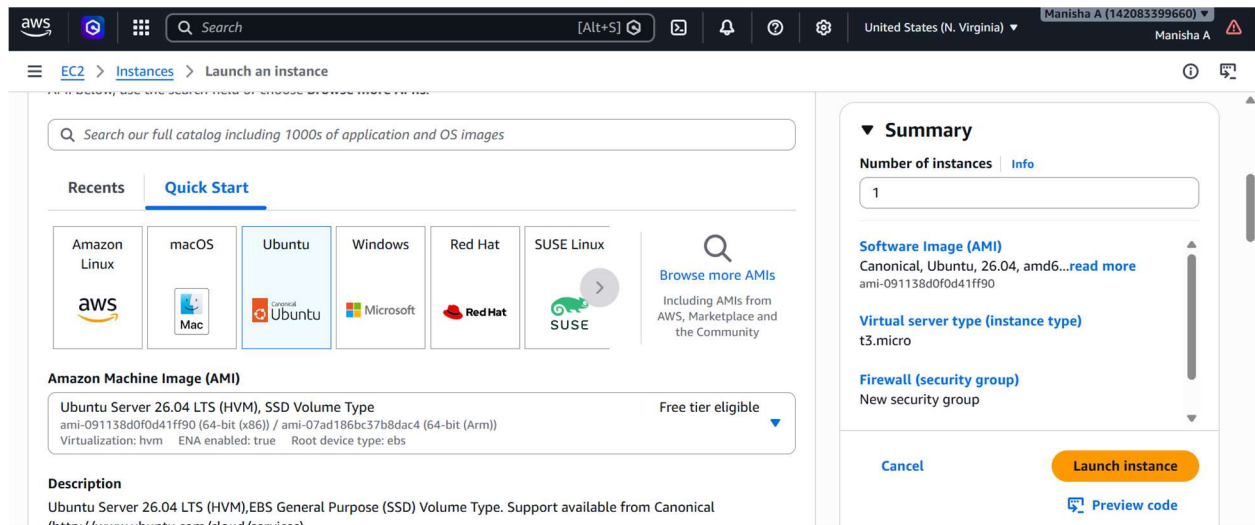
DOCKER-01

1. Setup a virtual machine using Ubuntu image.

First lets launch new instance



Select AMI as ubuntu



Choose instance type and key pair

aws

Search

[Alt+S]

United States (N. Virginia)

Manisha A (142083399660)

Manisha A

EC2 > Instances > Launch an instance

▼ Instance type Info | Get advice

Instance type

c7i-flex.large

Family: c7i-flex 2 vCPU 4 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.17219 USD per Hour

On-Demand RHEL base pricing: 0.11359 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.08829 USD per Hour

On-Demand Linux base pricing: 0.08479 USD per Hour

On-Demand SUSE base pricing: 0.14109 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

jen

Create new key pair

▼ Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd6...read more

ami-091138d0f0d41ff90

Virtual server type (instance type)

c7i-flex.large

Firewall (security group)

New security group

Cancel

Launch instance

Preview code

aws

Search

[Alt+S]

United States (N. Virginia)

Manisha A (142083399660)

Manisha A

EC2 > Instances > Launch an instance

Network Info

vpc-0e845fab348467a5b

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-24' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

▼ Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd6...read more

ami-091138d0f0d41ff90

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Cancel

Launch instance

Preview code

aws

Search

[Alt+S]

United States (N. Virginia)

Manisha A (142083399660)

Manisha A

EC2 > Instances > Launch an instance

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Configure storage Info Advanced

1x 20 GIB gp3

Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

▼ Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd6...read more

ami-091138d0f0d41ff90

Virtual server type (instance type)

c7i-flex.large

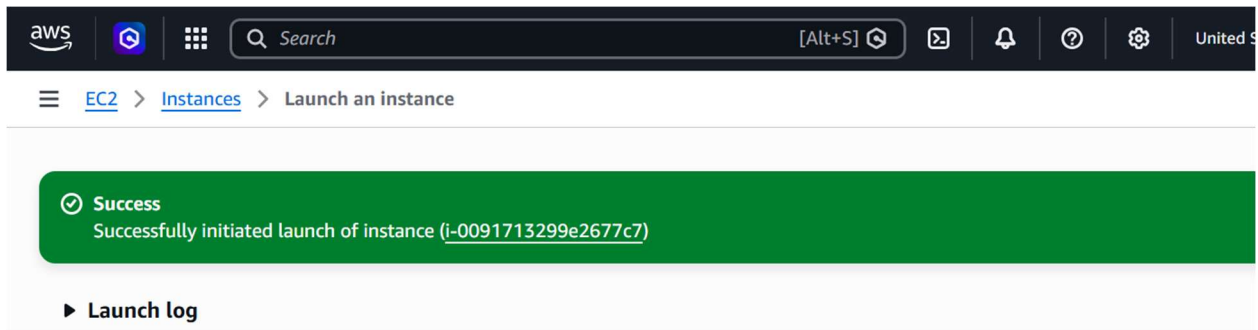
Firewall (security group)

New security group

Cancel

Launch instance

Preview code



Connect to instance

```
root@ip-172-31-24-106: /home/ubuntu
ubuntu@ip-172-31-24-106:~$ sudo su
root@ip-172-31-24-106:/home/ubuntu# sudo apt update -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-24-106:/home/ubuntu# sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  dns-root-data  pigz  ubuntu-fan
  containerd    dnsmasq-base   runc

Suggested packages:
  ifupdown          | cgroup-lite  docker-compose-v2  rootlesskit
  aufs-tools        debootstrap   docker-doc          zfs-fuse
  cgroupfs-mount    docker-buildx  rinse               | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 35
  Download size: 74.0 MB
```

2.Install docker on ec2.

```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  dns-root-data  pigz  ubuntu-fan
  containerd    dnsmasq-base   runc

Suggested packages:
  ifupdown          | cgroup-lite  docker-compose-v2  rootlesskit
  aufs-tools        | debootstrap  docker-doc          | zfs-fuse
  cgroupfs-mount    | docker-buildx  rinse              | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 35
  Download size: 74.0 MB
  Space needed: 281 MB / 17.2 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3
  pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 brid
  e-utils amd64 1.7.1-4ubuntu3 [34.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc
  amd64 1.4.0-0ubuntu1 [9829 kB]

Setting up docker.io (29.1.3-0ubuntu4.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' →
  /usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/us
  /lib/systemd/system/docker.socket'.
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-24-106:/home/ubuntu# sudo systemctl start docker
root@ip-172-31-24-106:/home/ubuntu# sudo systemctl enable docker
root@ip-172-31-24-106:/home/ubuntu# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
root@ip-172-31-24-106:/home/ubuntu#
```

3. Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

To pull docker images

For nginx – **docker pull nginx**


```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
13fd728be9eb: Pull complete
5b4d6ff92fc4: Pull complete
830625e1ac85: Pull complete
5431d0092ffd: Pull complete
b4a248c845e5: Pull complete
7f8b1a2b17d8: Pull complete
45381ecb0e2f: Pull complete
cc5f57206478: Download complete
5e6b66b5e5f1: Download complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

For tomcat- **docker pull tomcat**

```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
d92a694d8aa0: Pull complete
b40150c1c271: Pull complete
6b82a07ff6e7: Pull complete
97c47a5c1a2f: Pull complete
b38893372039: Pull complete
65313903a8a8: Pull complete
13883d0c1f97: Download complete
f80ef884f8c2: Download complete
Digest: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
```

Ubuntu : **docker pull ubuntu**

Jenkins : **docker pull jenkins/jenkins:lts**

```

root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
root@ip-172-31-24-106:/home/ubuntu# docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
307f8152a55e: Pull complete
6b35bb76b8c8: Pull complete
3bdbd616c406: Pull complete
cebc1e77ceba: Pull complete
5834ae4b8774: Pull complete
801814f62c99: Pull complete
c2314d42cca1: Pull complete
ae218d1cfa6e: Pull complete
76a355a69c2e: Pull complete
9014a3401c36: Pull complete
dd00b4ea1f64: Pull complete
4c140080734e: Pull complete
653a4e57bccf: Download complete

```

Sonarqube - **docker pull sonarqube**

```

root@ip-172-31-24-106:/home/ubuntu# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
c10b8232386e: Pull complete
f4306f81d85a: Pull complete
4f4fb700ef54: Pull complete
907ea4a8bc72: Download complete
8506693bc963: Download complete
Digest: sha256:35bedac3f40cda75969890da59b17d577770844fe6ef659206c678a8e00921c7
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
root@ip-172-31-24-106:/home/ubuntu#

```

To verify

docker images

```

root@ip-172-31-24-106:/home/ubuntu# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	
nginx:latest	5aca99593157	241MB	66MB	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```

root@ip-172-31-24-106:/home/ubuntu#

```

4. Run nginx container and expose on port 81.

Run container

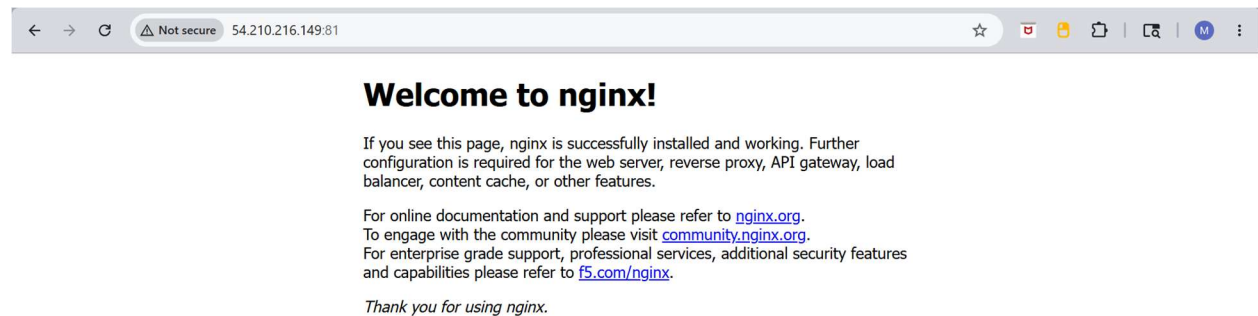
docker run -d --name nginx-container -p 81:80 nginx

```
root@ip-172-31-24-106:/home/ubuntu# docker run -d --name nginx-container -p 81:80 nginx
d0955fb435e63cd953ebc0425650cdf440986e60ae8129a8490b862487c69514
root@ip-172-31-24-106:/home/ubuntu# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
d0955fb435e6	nginx	"/docker-entrypoint..."	22 seconds ago	Up 21 seconds
0.0.0.0:81->80/tcp, [::]:81->80/tcp nginx-container				

```
root@ip-172-31-24-106:/home/ubuntu#
```

Verified browser and we are able to reach nginx page



5. Delete the Apache tomcat image from local.

First lets check docker images

```
root@ip-172-31-24-106:/home/ubuntu# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
jenkins/jenkins:lts	01c992ffef29	799MB	289MB		
nginx:latest	5aca99593157	241MB	66MB	U	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB		
tomcat:latest	247650787b97	589MB	158MB		
ubuntu:latest	f3d28607ddd7	160MB	45.3MB		

```
root@ip-172-31-24-106:/home/ubuntu#
```

Deleted tomcat image and verified

```
root@ip-172-31-24-106:/home/ubuntu# docker rmi -f tomcat
Untagged: tomcat:latest
Deleted: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
root@ip-172-31-24-106:/home/ubuntu# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Use
jenkins/jenkins:lts	01c992ffef29	799MB	289MB		
nginx:latest	5aca99593157	241MB	66MB	U	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB		
ubuntu:latest	f3d28607ddd7	160MB	45.3MB		

```
root@ip-172-31-24-106:/home/ubuntu#
```

6. Inspect the Jenkins image, SonarQube image.

Inspect Jenkins image

docker image inspect jenkins/jenkins:lts

```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker image inspect jenkins/jenkins:lts
[
  {
    "Id": "sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5",
    "RepoTags": [
      "jenkins/jenkins:lts"
    ],
    "RepoDigests": [
      "jenkins/jenkins@sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-13T17:19:33.598838948Z",
    "Config": {
      "User": "jenkins",
      "ExposedPorts": {
        "50000/tcp": {},
        "8080/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",

```

```

        "sha256:db80923e26ae1d830f7849e23fb76268a8acae9827ed137e99200a39a6e40810",
        "sha256:71d9981f81e463e428308b0fc6dc5fc85578612e750cca5a346f8a3b9b6c130f",
        "sha256:2e6d205ec218115bff7e4a4aba21b9e88e2bc8fad05fc86171e84f99a960c0ca",
        "sha256:2b04b11bf1853abb8be2ea3fdca158d8f82556aba2be08f5d428d5463b488daa",
        "sha256:a47bf6ef3ffe65239dec07e31f2908c0309f9f33deb864ccd23b2a6bc8592eff",
        "sha256:49ccb4bf770fc42dad334c2eb9fb5379a4f6b24ffa3db0fa81f03368d7de0f18"
      ]
    },
    "Metadata": {
      "LastTagTime": "2026-05-27T10:53:11.455167204Z"
    },
    "Descriptor": {
      "mediaType": "application/vnd.oci.image.index.v1+json",
      "digest": "sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5",
      "size": 3872
    }
  }
]
```

Inspect sonarqube image

docker image inspect sonarqube


```

root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker image inspect sonarqube
[
  {
    "Id": "sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7",
    "RepoTags": [
      "sonarqube:latest"
    ],
    "RepoDigests": [
      "sonarqube@sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-11T22:13:40.873390969Z",
    "Config": {
      "User": "sonarqube",
      "ExposedPorts": {
        "9000/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "JAVA_HOME=/opt/java/openjdk",
        "LANG=en_US.UTF-8"
      ]
    }
  }
]

```

```

root@ip-172-31-24-106: /home/ubuntu
    "sha256:5f70bf18a086007016e948b04aed3b82103a36bea41755b6cddfaf10ace3c6ef",
    "sha256:84d5dfc5418b25bf47784deaf4b0a4b95ff3a1347d126004d59ad2cf42e08fd7",
    "sha256:0a9af819d82fd697584bf5d5af4bbdf704a0f33636574d0db5f80edb6dad28cfc",
    "sha256:d92a086e51cb98d1f29cf3d1f4432ac06c7e454f3555bdc06f93d9924d24d3b1",
    "sha256:5f70bf18a086007016e948b04aed3b82103a36bea41755b6cddfaf10ace3c6ef"
  ],
  "Metadata": {
    "LastTagTime": "2026-05-27T10:53:36.998584325Z"
  },
  "Descriptor": {
    "mediaType": "application/vnd.oci.image.index.v1+json",
    "digest": "sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7",
    "size": 2707
  }
}
]
root@ip-172-31-24-106:/home/ubuntu#

```

7. Run Jenkins container and run one sample job.

Run Jenkins

docker run -d \

--name jenkins-container \

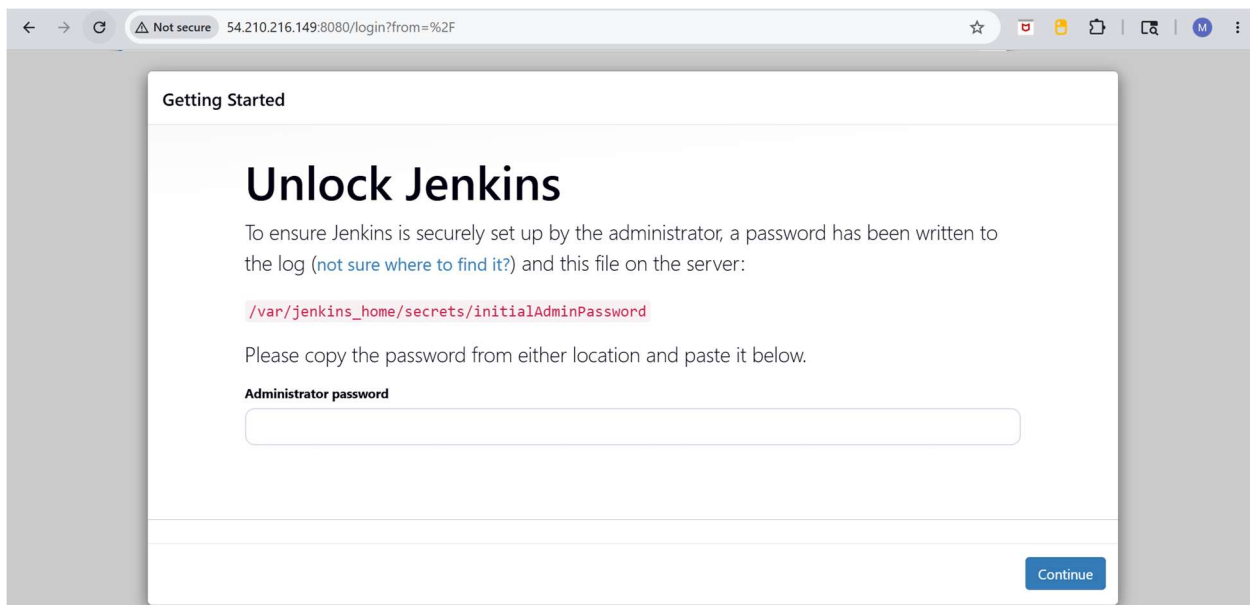
-p 8080:8080 \

-p 50000:50000 \

jenkins/jenkins:latest

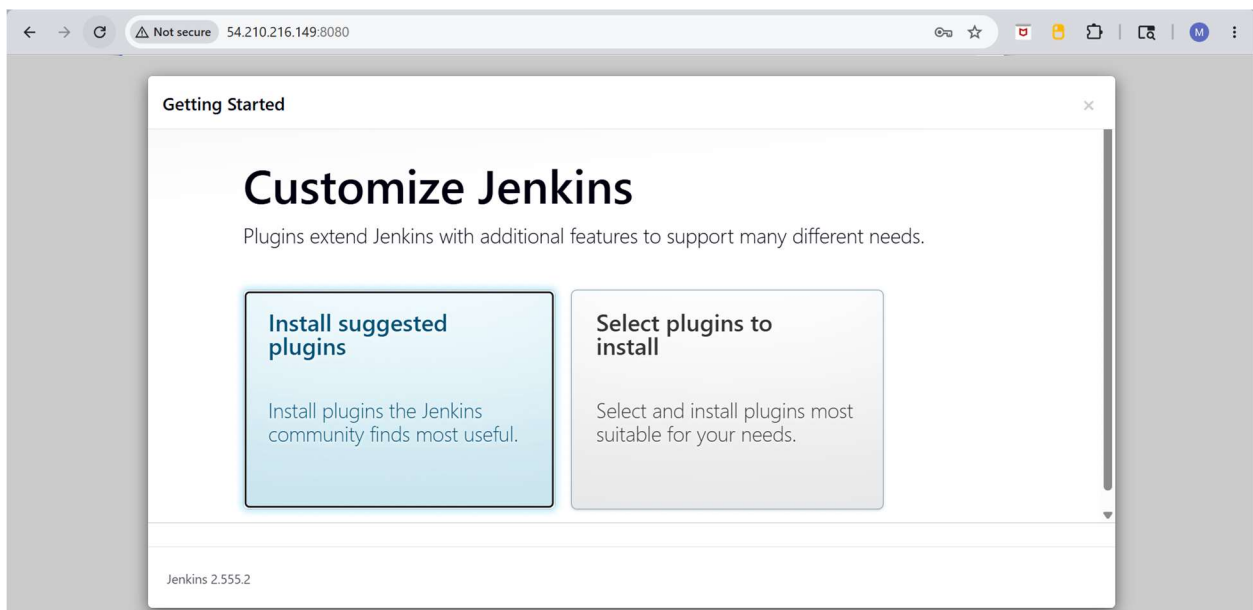
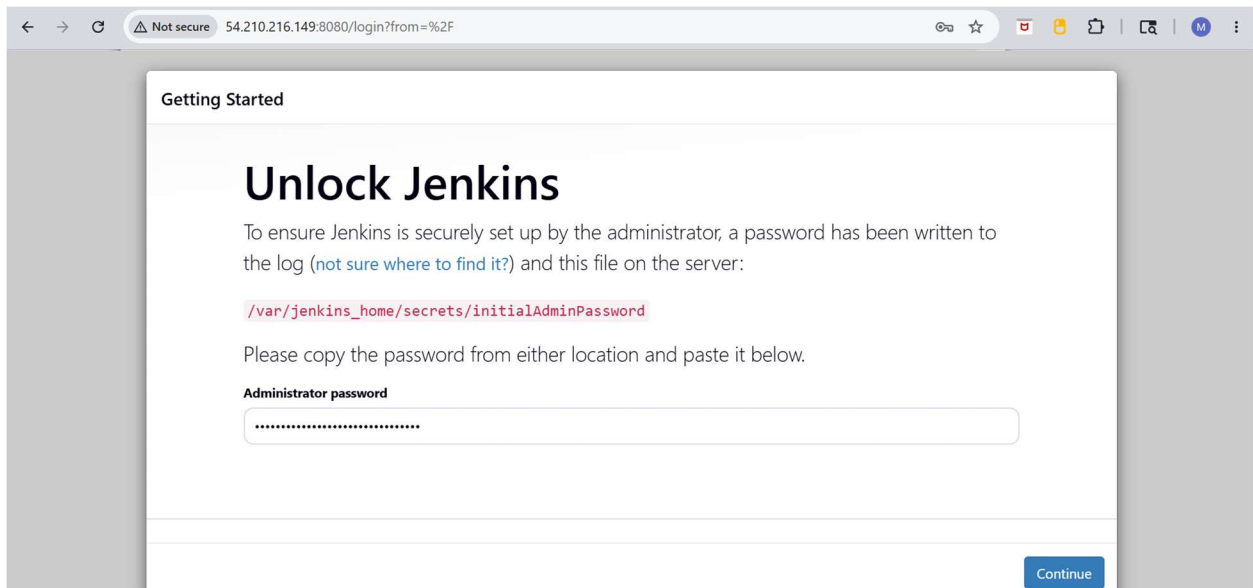
```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker run -d \
--name jenkins-container \
-p 8080:8080 \
-p 50000:50000 \
jenkins/jenkins:latest
4c2f9e0edd117aa7ce352b84477601b7287ddf517614f9e6c574af3ee06b38f0
root@ip-172-31-24-106:/home/ubuntu# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
4c2f9e0edd11   jenkins/jenkins:latest "/usr/bin/tini -- /u... 27 seconds ago Up
p 26 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000
/tcp, [::]:50000->50000/tcp   jenkins-container
d0955fb435e6   nginx         "/docker-entrypoint.... 14 minutes ago Up
p 14 minutes   0.0.0.0:81->80/tcp, [::]:81->80/tcp
nginx-container
root@ip-172-31-24-106:/home/ubuntu#
```

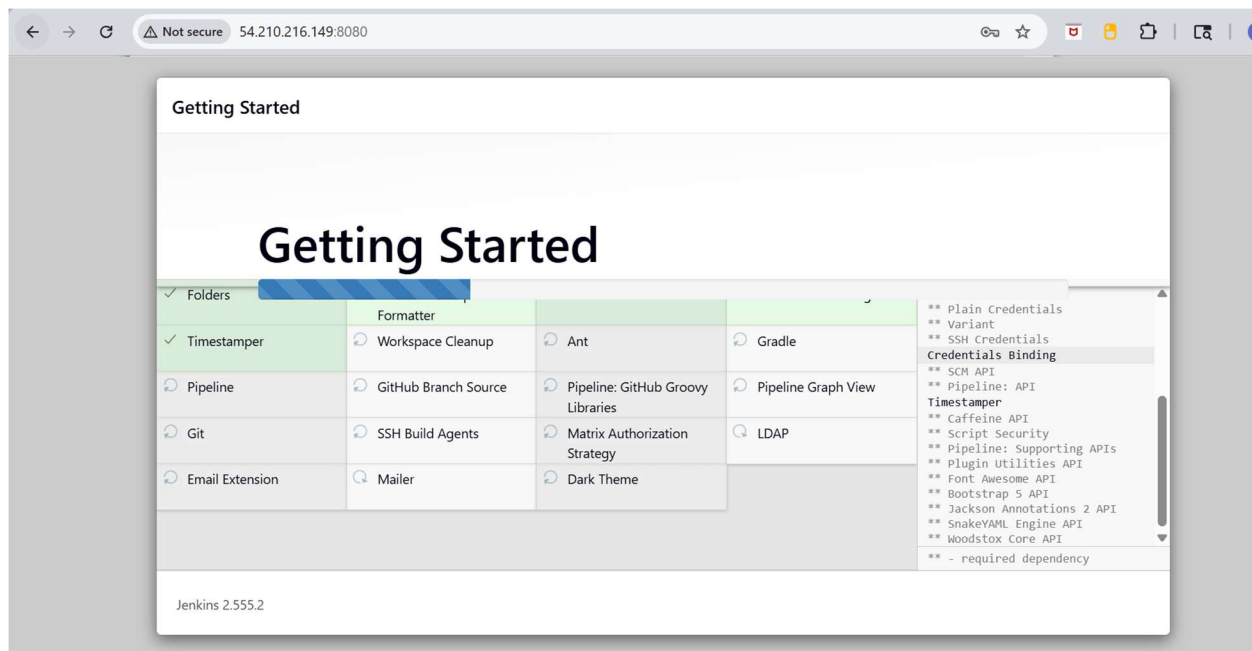
Access Jenkins



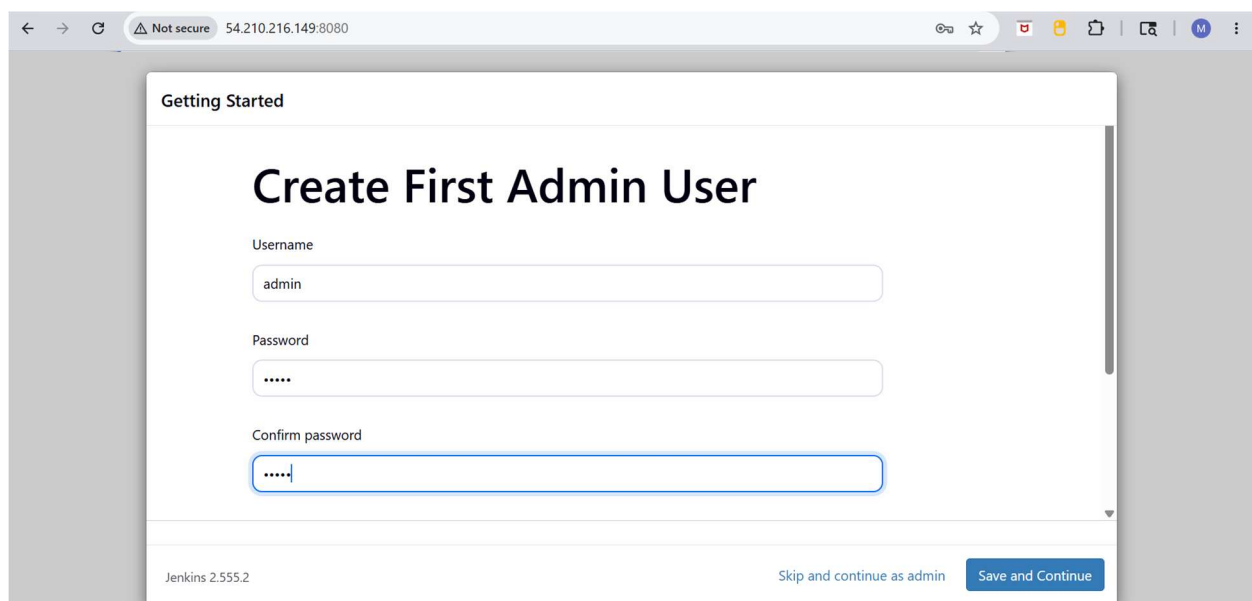
To get Jenkins Initial Password

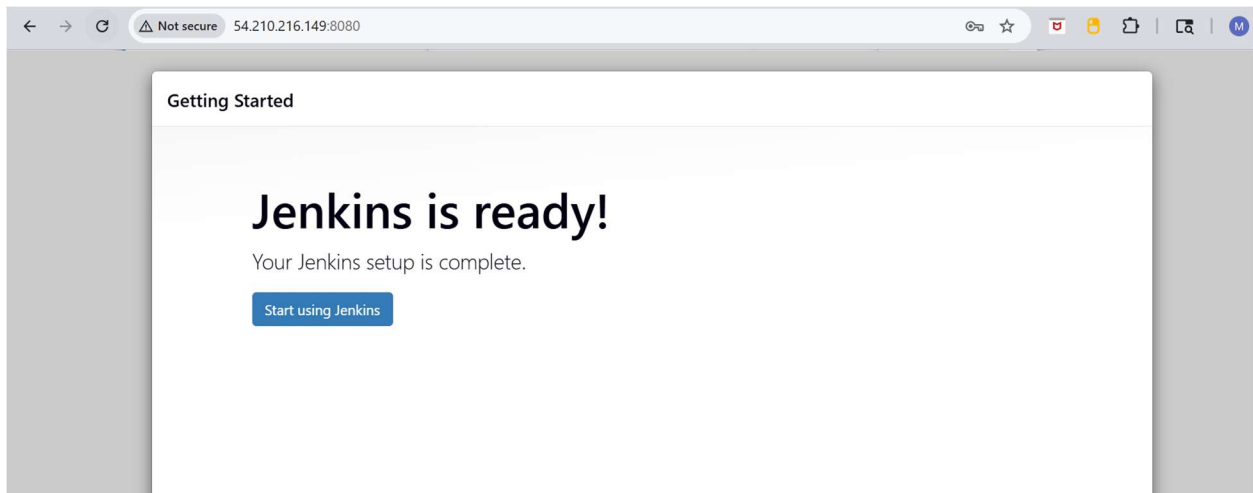
```
docker exec jenkins-container cat /var/jenkins_home/secrets/initialAdminPassword
```



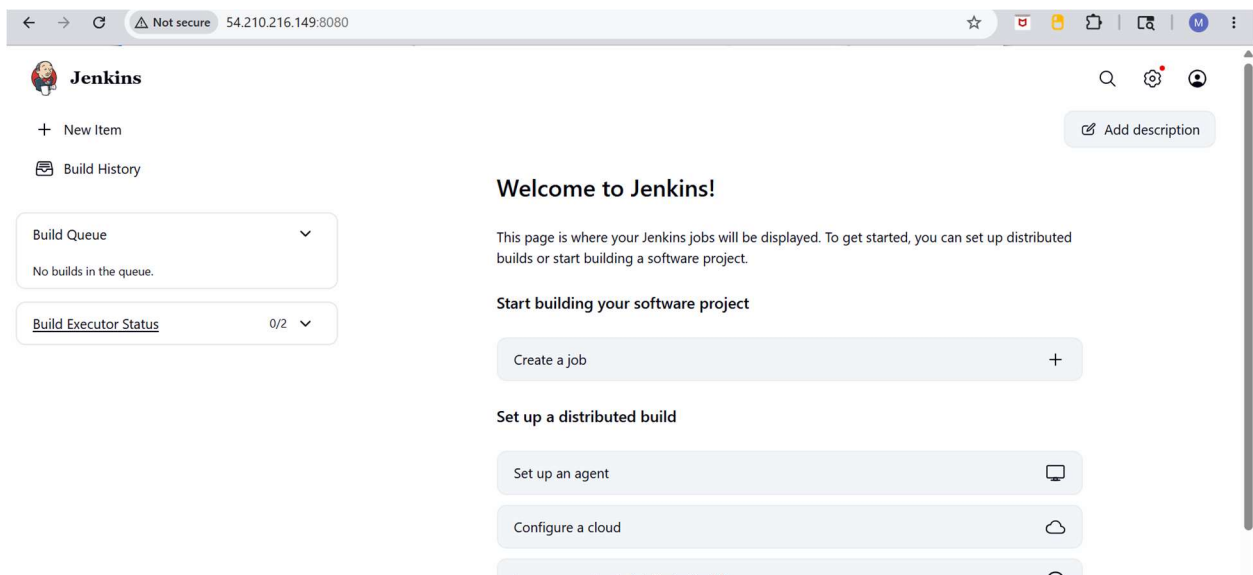


Create user and password





Now lets create a new build



Choose freestyle

← → ↻ ⚠ Not secure 54.210.216.149:8080/view/all/newJob ☆ 📄 🗑 📄 📄 📄

 **Jenkins** / All ▾ / New Item 🔍 ⚙ 👤

New Item

Enter an item name

job-1

Select an item type

 **Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Choose execute shell

← → ↻ ⚠ Not secure 54.210.216.149:8080/job/job-1/configure ☆ 📄 🗑 📄 📄 📄

 **Jenkins** / job-1 ▾ / Configure 🔍 ⚙ 👤

Configure

 General

 Source Code Management

 Triggers

 Environment

 **Build Steps**

 Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

⋮ Execute shell ? ✕

Command

[See the list of available environment variables](#)

```
echo "hello world"
```

Advanced ▾

Save Apply

Click on build now to run the job

← → ↻ ⚠ Not secure 54.210.216.149:8080/job/job-1/ ☆ 📄 📁 🔍 | M ⋮

 **Jenkins** / job-1 🔍 ⚙️ 👤

Status

</> Changes

📁 Workspace

▶ Build Now

⚙️ Configure

🗑 Delete Project

✎ Rename

job-1
job-1
Permalinks

Edit description

Builds > ⋮

Today
✔ #1 11:29 am ▼

REST API Jenkins 2.555.2

← → ↻ ⚠ Not secure 54.210.216.149:8080/job/job-1/1/ ☆

 **Jenkins** / job-1 ▼ / #1 ▼ 🔍 ⚙️ 👤

Status

</> Changes

📄 Console Output

✓ Edit Build Information

🗑 Delete build '#1'

🕒 Timings

✔ #1 (27-May-2026, 11:29:38 am)

🕒 Started by user [admin](#)

🕒 This run spent:

- 31 ms waiting;
- 47 ms build duration;
- 78 ms total from scheduled to completion.

</> No changes.

Add description

Check the console output

← → ↻ ⚠ Not secure 54.210.216.149:8080/job/job-1/console ☆ 📄 📁 🔍 | M ⋮

 **Jenkins** / job-1 ▼ / #1 ▼ 🔍 ⚙️ 👤

Status

</> Changes

📄 Console Output

✓ Edit Build Information

🗑 Delete build '#1'

🕒 Timings

Console

Download Copy View as plain text

Started by user [admin](#)
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/job-1
[job-1] \$ /bin/sh -xe /tmp/jenkins3082197602289684907.sh
+ echo hello world
hello world
Finished: SUCCESS

